

Marwan Abdelaal

Backend and Full-Stack Developer | AI Engineering Intern | Software Engineering Student

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SUMMARY

2nd-year Software Engineering student at GIU (GPA: 2.4, Very Good, German Scale) with a Minor in IT Security. Currently interning as a Backend AI Engineering Intern at FlyRank AI, where I have shipped production-ready REST APIs, persistent database systems, JWT authentication, web scraping pipelines with local AI enrichment, and automated test suites with 100% pass rates. Proficient in Node.js, Express.js, Python, SQL, PostgreSQL, Docker, and React.js, with hands-on experience in AI integration, cryptography, distributed systems, machine learning, and prompt engineering. Led a team of 10 to ship an AI-powered MERN platform. Holder of 20 Anthropic Academy certifications.

SKILLS

Backend:	Node.js Express.js REST APIs JWT Middleware Supabase Auth Swagger UI
Languages:	JavaScript Python Java C# SQL HTML and CSS
Databases:	SQLite PostgreSQL MongoDB Microsoft SQL Server Schema Design Query Optimization
AI and ML:	Hugging Face Ollama NLP Prompt Engineering Embeddings Naive Bayes KNN K-Means PCA
Security:	3DES RSA Diffie-Hellman AES Cryptographic Protocols SQL Injection Prevention
Data and Scraping:	Cheerio Axios Zod pandas scikit-learn matplotlib Jupyter Notebook
DevOps:	Docker Docker Compose Git GitHub GitLab CI/CD
Frameworks:	React.js JavaFX Unity .NET
Distributed:	MapReduce Weak Consistency Distributed Locking Synchronization
Testing:	Automated Testing Postman Manual Test Cases Parameterized Queries API Validation
Soft Skills:	Leadership Communication Problem-Solving Agile Scrum Time Management

PROFESSIONAL EXPERIENCE

Backend AI Engineering Intern - FlyRank AI	Jun 2026 - Present Remote
* Built a production-ready Node.js/Express REST API with 12 endpoints, full CRUD, SQLite persistent storage, parameterized queries for SQL injection prevention, and 23 automated tests all passing with 100% success rate. * Implemented a secure JWT authentication system using Supabase Auth with reusable middleware, Swagger UI documentation, and proper 401/403 HTTP status distinction for protected route management. * Developed a production-grade web scraping pipeline collecting 60 books using Cheerio and Axios with Zod schema validation, exponential backoff retry logic, caching reducing re-run time from 45s to 1s, CSV export, and local AI enrichment via Ollama. * Migrated backend storage to PostgreSQL in Docker using a repository pattern that swaps storage without changing service or route logic, orchestrated via Docker Compose with proven data persistence across restarts. * Engineered a 5-iteration prompt ladder applying role assignment, context framing, few-shot examples, and output structure techniques, comparing outputs across Claude and ChatGPT to produce reusable prompt templates. * Completed 20 Anthropic Academy certifications covering AI fluency, Claude tooling, MCP, agent skills, subagents, Claude API, Amazon Bedrock, and Google Cloud integrations.	

Part-Time Teaching Assistant - ICTeam El Komy	Jun 2024 - May 2026 Cairo, Egypt
* Delivered technical guidance to 20+ students across lab sessions, simplifying complex concepts and troubleshooting issues in real time, strengthening communication and leadership skills. * Adopted an agile approach to adapt session content based on student feedback, ensuring continuous improvement across multiple class cycles.	

Event Volunteer and Usher - Various Organizations	2024 - Present Cairo, Egypt
* Coordinated operations across 3+ large-scale events including the Cairo International Film Festival and university career fairs, managing volunteer teams under high-pressure conditions.	

PROJECTS

GIU Nexus - AI-Powered Career and Talent Platform
<i>Tech: React Vite, Node.js, Express.js, MongoDB, Hugging Face API, JWT, Nodemailer, GitLab github.com/MoscowAbdelaal/GIU-Nexus</i>
* Led a team of 10 to build a full-stack MERN application with Hugging Face NLP for skill extraction and zero-shot classification, cosine similarity recommendations, JWT auth, and webhook-based notifications.

* Optimized backend performance by improving query efficiency, reducing API response times, and validated all endpoints via Postman across functional, integration, and regression test scenarios.

CRUD API - Production REST API with Auth and Testing

Tech: Node.js, Express.js, SQLite, PostgreSQL, Docker, Supabase Auth, JWT, Swagger UI | github.com/MoscowAbdelaal/crud-api

* Built a full REST API with 12 endpoints, persistent SQLite storage, 23 automated tests at 100% pass rate, JWT auth via Supabase, reusable middleware, and Swagger UI docs, then migrated to PostgreSQL in Docker using a clean repository pattern.

Clash of Clans: Heroes - 2-Player Strategy Game

Tech: Java, JavaFX, Eclipse IDE, OOP, CSV Data Loading | github.com/MoscowAbdelaal/Clash-of-clans-Legends

* Independently built a complete 2-player turn-based strategy game across 3 milestones with full OOP architecture including abstract classes, interfaces, inheritance, custom exception handling, and a JavaFX frontend with real-time updates.

Human Resource Management System (HRMS)

Tech: Microsoft SQL Server, HTML, CSS, Node.js, SQL Functions and Stored Procedures

* Led full EERD and relational schema design, implemented optimized SQL stored procedures as the backend layer, and delivered a multi-role system covering payroll, attendance, leave, hiring, and performance reviews.

Web Scraping Pipeline with AI Enrichment

Tech: Node.js, Cheerio, Axios, Zod, Ollama, HTTP Server

* Production-grade scraper collecting 60 books with Zod validation, exponential backoff, polite rate limiting, caching from 45s to 1s, CSV export, change detection, and local AI enrichment via Ollama for category and summary generation.

Cryptography Suite - 3DES and Diffie-Hellman/RSA

Tech: Python, Jupyter Notebook

* Implemented Triple DES encryption from scratch using EDE structure across 16-round blocks, and built a secure communication system combining Diffie-Hellman key exchange with RSA digital signatures simulating HTTPS-like secure transmission.

Machine Learning - Classification and Clustering

Tech: Python, scikit-learn, pandas, matplotlib, Jupyter Notebook, Kaggle

* Built Naive Bayes and KNN classifiers to predict metabolic syndrome risk with hyperparameter tuning and Kaggle leaderboard submission. Applied K-Means, GMM, BIRCH, and Agglomerative Clustering to 561-feature HAR sensor data with PCA visualization and anomaly detection.

Distributed Systems - MapReduce and Theater Booking Simulation

Tech: Python, Google Colab

* Implemented MapReduce for inverted index, temperature aggregation, movie ratings, and product sales. Built a distributed theater booking simulation with 4 parallel counters, weak consistency, local queuing, and synchronized global state across 5 rounds using distributed locking.

Interactive Media Experience - Unity

Tech: Unity, C#, AudioSource, Animator, TextMeshPro

* Designed and built an interactive Unity experience with 4+ media types including graphics, animation, audio, and text, featuring 3+ meaningful user interactions, state-based media responses, and a fully playable build.

CERTIFICATIONS - ANTHROPIC ACADEMY (2026)

* AI Fluency: Framework and Foundations	* Claude 101
* Claude Code 101	* Claude Code in Action
* Claude Platform 101	* AI Capabilities and Limitations
* AI Fluency for Students	* AI Fluency for Educators
* AI Fluency for Builders	* AI Fluency for Small Businesses
* AI Fluency for Nonprofits	* Teaching AI Fluency
* Introduction to Claude Cowork	* Introduction to Agent Skills
* Introduction to Subagents	* Introduction to Model Context Protocol
* Model Context Protocol: Advanced Topics	* Building with the Claude API
* Claude with Amazon Bedrock	* Claude on Google Cloud

EDUCATION

**Bachelor of Software Engineering - Informatics and Computer Science -
German International University (GIU)**

Sep 2024 - Present | Cairo, Egypt

GPA: 2.4 / Very Good (German Scale) | Minor: IT Security | Expected Graduation: 2028 | Coursework: Data Structures and Algorithms, OOP, Database Systems, Data Science, Information Security, Distributed Systems, Media Informatics

LANGUAGES

Arabic (Native) | English (Fluent) | German (Beginner) | French (Beginner)

HOBBIES

Traveling | Swimming | Music | Fitness and Gym | Motorcycle Riding | Diving